

Lab 8 - Work and Energy

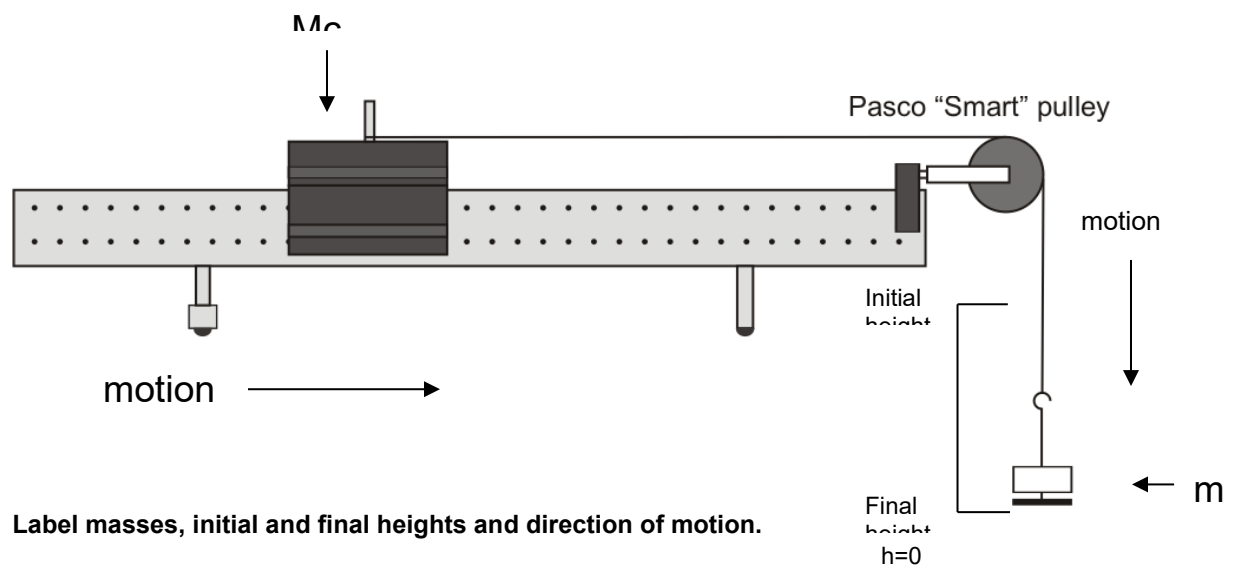
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Learning Outcomes

1. Use work-energy relationships to make predictions.
2. Integrate multiple physics concepts to solve a variety of problems
3. Create and perform experiments that make experimental determinations and predictions of unknown quantities.

Materials and Apparatus

- Laptop
- Pasco Capstone software
- Smart Gate
- “frictionless” pulley
- air track
- air cart
- hanging mass
- string
- red spring



Experiment 1 - Work and Energy

An air cart of mass M_c is tied to a mass m by a string that hangs over a light pulley of negligible friction as shown. When M_c is at the left end of the track, m is at height h . When M_c is at the right end of the track, m is on the floor at $h = 0$.

Pre-lab theory

Use the **Work-Energy Theorem** to predict a numerical value for the **final speed** v_f of the cart of mass M_c after the hanging mass m falls a distance h .

1. First, calculate the work done by gravity on m ($W = F_g d \cos \theta$).

$$W_g = mgh$$

2. Then, determine the change in kinetic energy for the system (ΔE_k).

$$\Delta KE = \frac{1}{2}(M_c + m)v^2$$

3. Use the equations to solve for v^2 in terms of m , g , h and M_c .

$$v^2 = \frac{2mgh}{M_c + m}$$

4. Take the square root to solve for v in terms of m , g , h and M_c .

$$v = \sqrt{\frac{2mgh}{M_c + m}}$$

5. You now have an equation for v to use to predict final speed **after** you setup the apparatus and take measurements of m , M_c and h . What units should you expect?

$$[v] = \sqrt{\frac{2(Kg)(m/s^2)(m)}{Kg + Kg}} = m/s$$

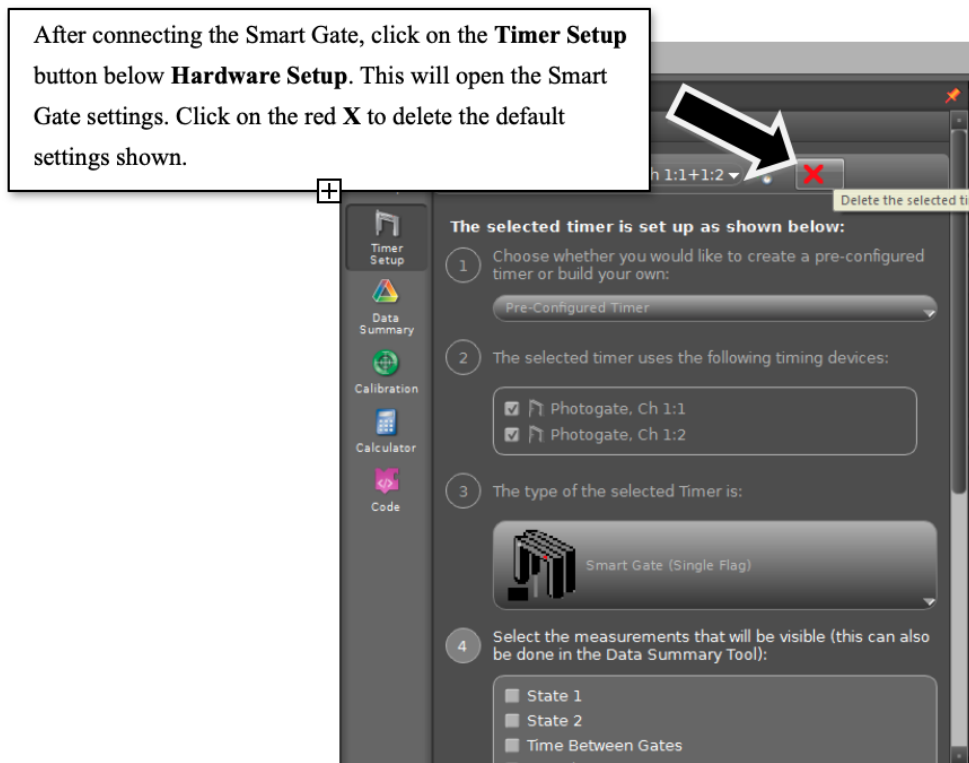
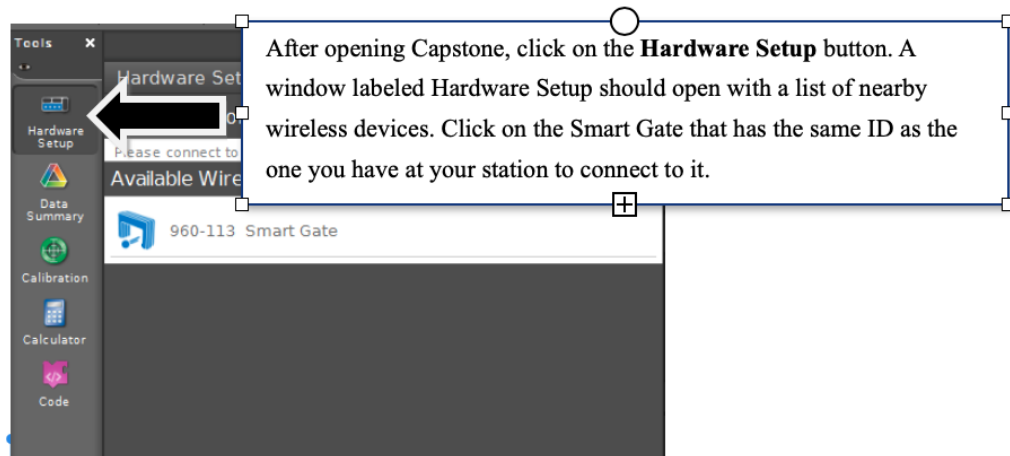
Setup apparatus & calculate predicted speed

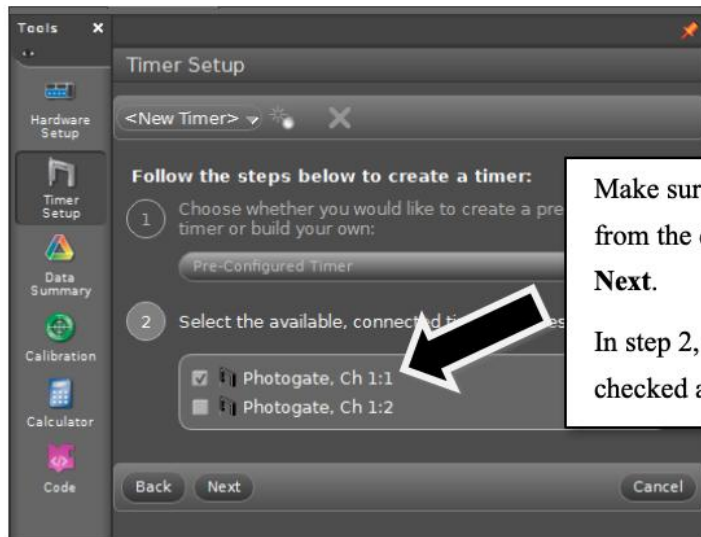
- After making your prediction, **mass** the cart and hanging mass on the digital scale. **Record** your measurements here, as measured and in SI units.
- **Set up** the air track, cart, pulley, Smart Gate and hanging mass m as shown in the figure and described above.

- Position and hold the cart at the left end of the track and measure ***h***. Record your measurement here, as measured and in SI units.
- Calculate the predicted value of final speed using your prelab equation and SI units.

Configure Capstone for linear speed measurements

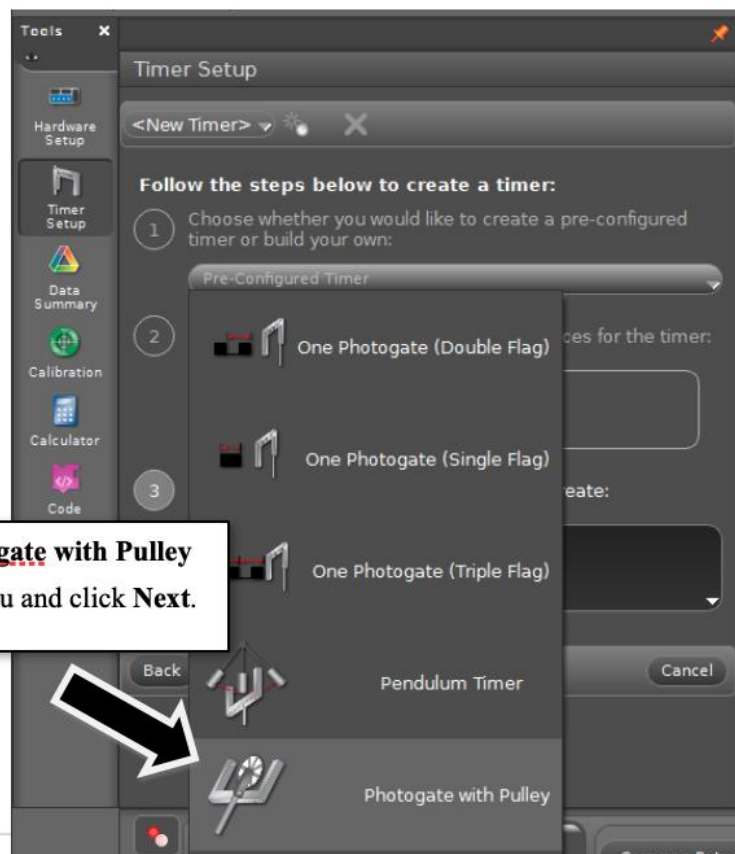
- **Press** the power button on the wireless Smart Gate to turn it on. This sensor will measure the ***angular speed ω*** (in radians per second) of the pulley as it rotates while the mass is falling.
- **Configure** Capstone to measure the ***linear speed v*** (in m/s) of the cart from the angular speed of the pulley using the following screenshots and instructions.



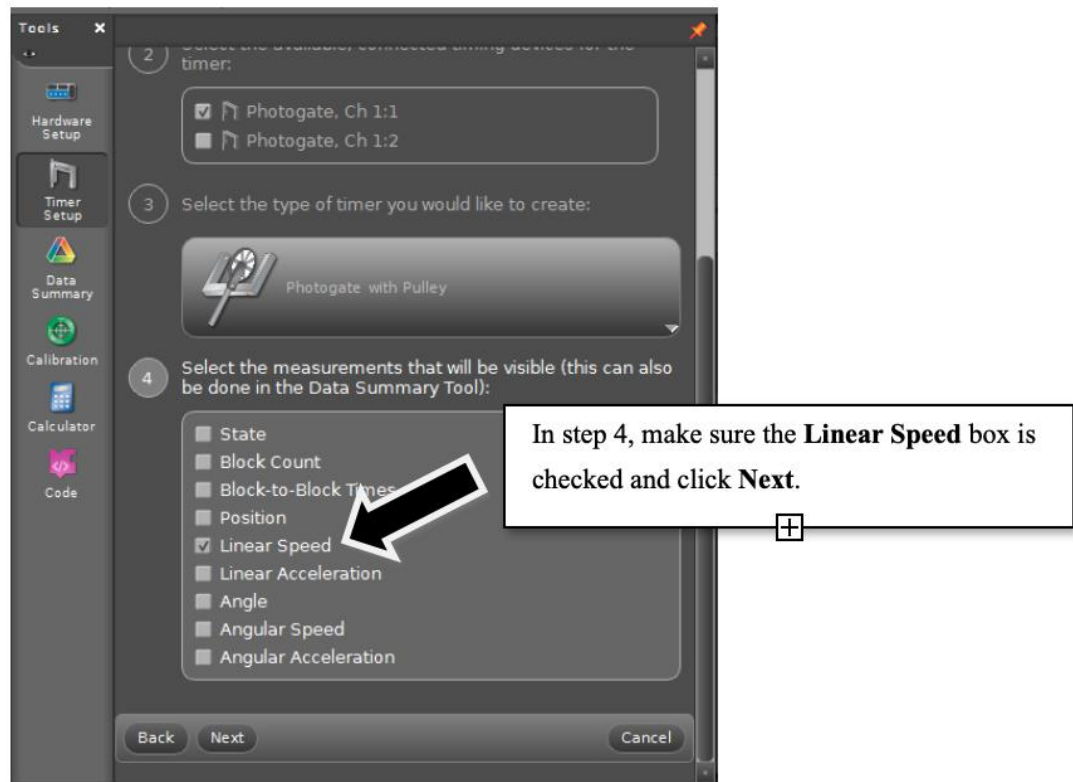


Make sure **Pre-Configured Timer** is selected from the drop-down menu in step 1 and click **Next**.

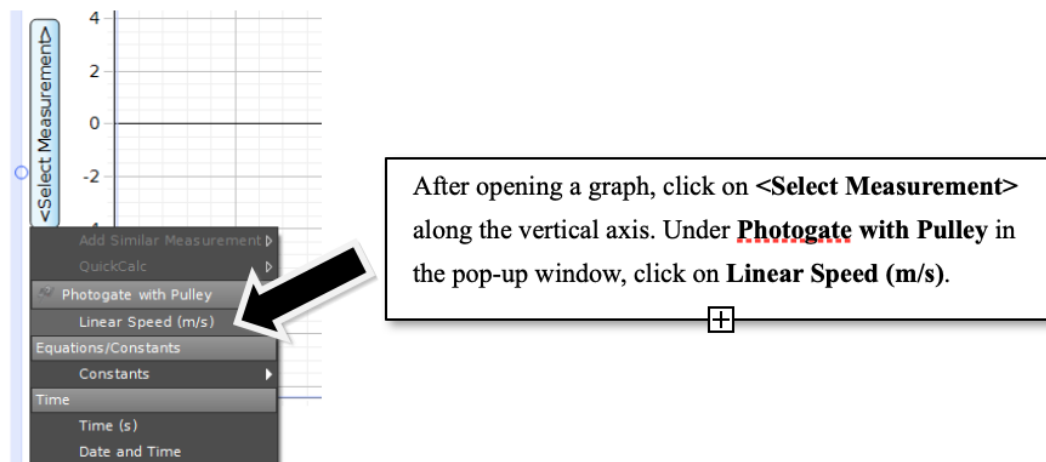
In step 2, only leave the **Photogate, Ch 1:1** box checked and click **Next** again.



In step 3, click on **Photogate with Pulley** from the drop-down menu and click **Next**.



- Continue clicking **Next** for the next several steps and then **Finish**.
- Click on the **Timer Setup** button to close the Timer Setup window. The Smart Gate is now configured to measure the linear speed of the cart as it moves down the air track.
- Double-click on the **Graph** icon to open a new graph.



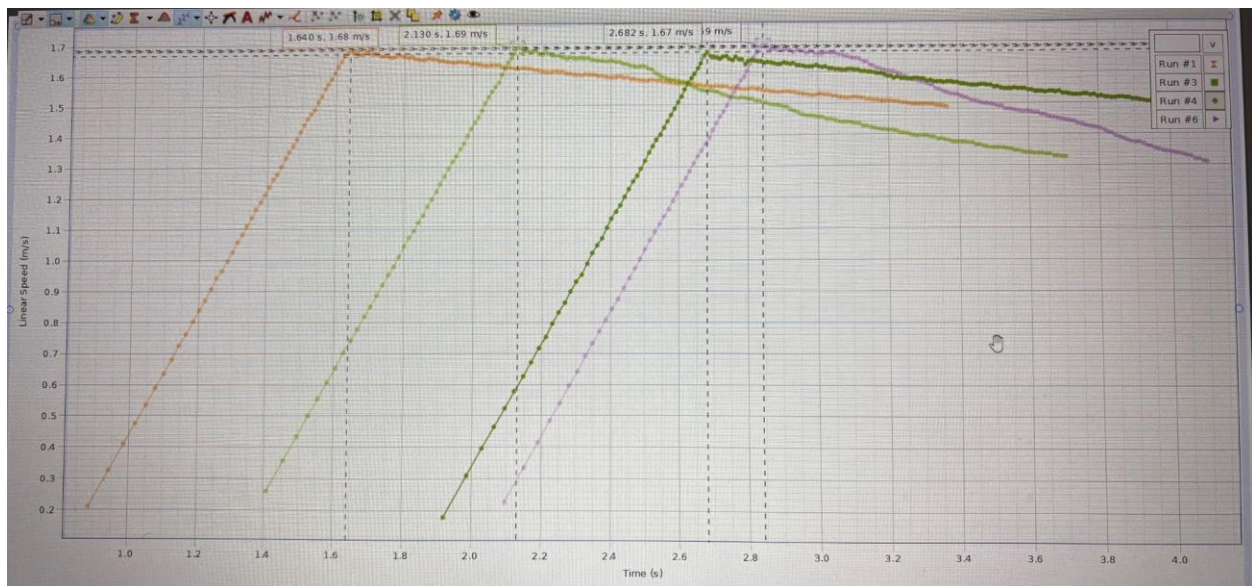
Select time (s) for the horizontal axis.

Measurement and comparison

PREDICTED VALUE: $v = 1.731327904$
 $= 1.73 \text{ m/s}$

- Position the cart near the far end of the track, click the **Record** button and turn the air track blower on. Using the linear speed vs. time graph, determine the final speed v_f of the cart after the hanging mass m falls a distance h to the floor.

Insert a screenshot of your graph and explain how you determined v_f .



We determined v final by finding the linear value of the peak. This is because this is the max velocity the cart reaches before the hanging mass hits the ground stopping acceleration.

- Compare your predicted result for the speed of the air cart to its experimentally measured value (calculate a percent error or percent difference as appropriate).

a. How closely do these results compare?

$$(1.734 - 1.6834) / 1.734 \times 100\% = 2.97\%$$

b. Account for any differences between these results.

Sources of difference for this include the fact that the equation measures if the hanging mass can reach terminal velocity, but the mass hits the ground before that so the cart never feels that max velocity. Error also comes from the fact that capstone was unable to zero the values for this data input.

- Which method would be easier to use to find the speed of the cart: the Work-Energy Theorem or Newton's 2nd Law? Explain by comparing steps used in this lab to steps used in Newton's 2nd Law lab or a problem done in class.

a. Do you think that both methods would give you similar results?

Both methods would give a similar result because they come from the same fundamental laws of physics. The Work-Energy Theorem. This theorem depends on the initial and final velocities vs. Newton's Second Law using mass and acceleration. Both can be found using similar data. For this lab, we used the Work-Energy Theorem more as we were working with the changes in potential and kinetic energy

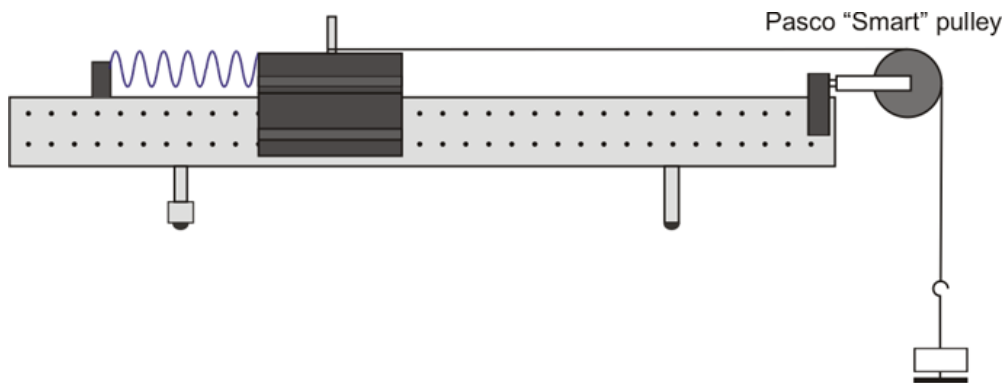
b. Why are you able to ignore the work done by the string when using conservation of energy for the whole system of the hanging mass and the air cart?

This is because the Net Work is zero because the sum of the forces in the y direction for the hanging mass = 0. The tension and the gravity are equal, which means that once you know one, you can find the other with ease. The string does positive work towards the rolling mass and negative work towards the hanging mass. Equal but opposite.

Experiment 2 - Conservation of Energy

A spring having elastic constant k is attached to one end of the air cart of mass M_c . The other end of the air cart is tied to a mass m by a string that hangs over a light pulley of negligible friction. The air cart is released from rest when the spring is at its **relaxed** length (x_0).

You may need to shorten the string so that when the spring is at its most extended position (x_0), the hanging mass m does not reach the floor at $h = 0$.



Background and prediction

Use conservation of energy ($\text{Total } E_{\text{initial}} = \text{Total } E_{\text{final}}$) to predict the maximum distance the air cart will stretch the spring (Δx). Express your answer in terms of k , the spring constant, and m , the hanging mass.

1. First, use an equation to represent the Total Energy of the system when the cart is at its relaxed length ($\text{Total } E_{\text{initial}}$).
2. Then, use another equation to represent the Total Energy of the system when the spring is at its maximum stretch (x_f).

Measurements

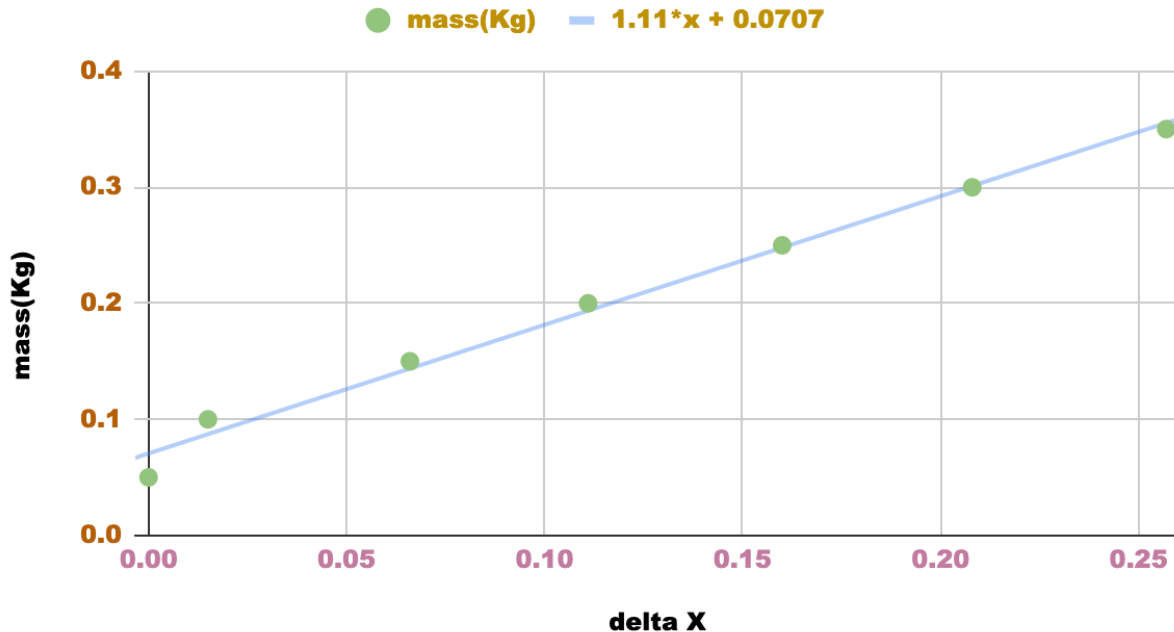
Add slotted masses to the hanging mass and measure the maximum extension of the spring for 10 different masses (when the spring is oscillating).

Record your results in a spreadsheet - practice making your own.

mass(Kg)	F _s	X (for spring)(m)	delta X	X (for mass)(m)
0.05	490.5	0.204	0	0.536
0.1	981	0.219	0.015	0.521
0.15	1471.5	0.27	0.066	0.47
0.2	1962	0.315	0.111	0.425
0.25	2452.5	0.364	0.16	0.374
0.3	2943	0.412	0.208	0.328
0.35	3433.5	0.461	0.257	0.279

Plot your results by making a scatter plot of

delta X vs. mass(Kg)



Based on the line of best fit, what is the value of k, the spring constant? Explain.

$$K = 21.7782$$

This is because the slope of mass(kg) vs delta x is $K/2g$, so we multiplied the 1.11 by 2 and 9.81 to get K

If there is time, your instructor will

Use a force sensor to measure the spring constant. Explain how to do this.

1.15

Compare your measurements of the spring constant (with a percent error or percent difference as appropriate). Do they agree?

$$\% \text{ difference} = \frac{1.15 - 1.11}{1.15} \times 100\% = 3.4\%$$

Which measurement do you trust more? Why?

We trust the force sensor more to measure the spring constant as it factors out all external forces that may cause uncertainty, compared to testing spring constants within an experiment